

Chapter 17: Effects of Light

Textbook Exercise Solutions & Answers



1. Fill in the blanks.

(a) When the beams from the headlights of a car fall on an object in the night, the shadows – called and can be seen.

umbra, penumbra

(b) During a lunar eclipse the shadow of the falls on the

earth, moon

(c) During a solar eclipse the shadow of the falls on the

moon, earth

(d) Various shades of colour are seen in the sky at sunrise and sunset due to

scattering of light

2. Give reasons.

(a) Space beyond the earth's atmosphere appears dark.

1. Space beyond the earth's atmosphere does have some gas and cosmic dust but there is not any atmosphere.
2. As there are no particles to scatter the sunlight, the space appears black.

(b) We are able to read while sitting in the shade.

1. We are able to read because the sun light which falls on the book is scattered and reaches our eyes.
2. While sitting in the shade our eyes adjust to the environment and to amount of light available. That is how we are able to read.

(c) We should not observe the solar eclipse with naked eyes.

1. During a solar eclipse ultra-violet rays which are harmful to us reach the earth and may lead to eclipse blindness or retinal burns and cataracts.
2. In order to protect our eyes a solar eclipse should never be watched with the naked eye.
3. A special type of goggles should be used for this.

3. Give some examples of scattering of light that we come across in day to day life.

1. The formation of rainbow, shift in position of stars, increased day time, mirage, inverted image, glittering of diamond, the working of lens and prism, bending of pen in water are examples of scattering of light.
2. The blue colour of the sky is due to the scattering of sunlight by the molecules of air.
3. During sunrise and sunset, sunlight has to travel greater distance, so shorter wavelength gets scattered off and removed and only orange and red with longer wavelengths reach us.
4. So during sunrise and sunset, sky appears red and orange.

4. Why is the shadow of a bird flying high not seen on the earth?

1. Birds flying high in the sky do cast their shadow but because they are shading an area that is very tiny the shadow is not visible.
2. The higher the bird flies, the smaller the shadow it casts.
3. Also when the bird flies high, the dark part of the shadow that is called Umbra does not reach the ground so we do not see its shadow.

5. Why is a penumbra not obtained from a point source?

1. The umbra, penumbra are the distinct parts of a shadow, created by any light source after striking on an opaque object.
2. For a point source, only the umbra is cast sharp dark shadow and not penumbra, because all the light of the point source will be blocked by any shadowing object.
3. Whereas penumbra forms only when some of the light from the source gets blocked by the shadowing object and not all of it does.

6. Answer the following questions in your own words.

(a) What is meant by scattering of light?

1. Scattering of light is the deviation of light rays from its straight path.
2. As light propagates through the atmosphere, it travels in a straight path until it is obstructed by bits of dust or gas molecules in the atmosphere
3. The process in which light gets deflected by the particles in the medium through which the light passes is called scattering, e.g. The blue colour of the sky is due to the scattering of sunlight by the molecules of air.

(b) Does the shadow really vanish in the zero shadow condition?

1. Yes, the day on which the sun reaches exactly overhead, at noon, shadow completely disappears. 'This event can be seen in the region between the Tropic of Cancer (23.5°N) and Tropic of Capricorn (23.5°S).
2. The shadow diminishes and eventually disappears for a while only to reappear later.

(c) Will the laser beam be seen if it passes through a glass box which contains a lighted incense stick?

Yes, it will be seen.

7. Discuss and write:

(a) Write a science based paragraph on 'What if the sun did not rise'?

1. The sun is a star and the centre of our solar system. Everything in our solar system revolves around the sun.
2. If the sun were to suddenly disappear, Earth and the other planets would retain their forward motion, effectively flying off into outer space in a straight line.
3. If the Sun didn't rise means the Earth's rotation had come to a screeching halt.
4. Sunrise and sunset are a result of the earth's rotation so we will not get to see sunrise or sunset. Earth's spinning generates the magnetic field at the core and it is saving us from harmful rays from the sun.
5. Without sun it would be very dark. No sun means no plants and no animals. Of course, without sun none of us would even exist.
6. Moon will disappear, because the moon does not produce light. We only see the moon because sunlight is reflected by the moon.
7. Without the sun's warmth, Earth would quickly become a much colder place. Life would be difficult, oceans will freeze.
8. Without sun rays, all photosynthesis on earth would stop. All plants would die.
9. All animals that rely on plants for food, including humans would die.

(b) What efforts will you make to remove the misconceptions about eclipses?

Solar, lunar eclipse have been traditionally observed as an ominous sign and therefore superstitions are prevalent in association with these natural phenomena.

1. All these misconceptions should be removed by explaining scientifically the process of eclipse how it happens: (a) Explain with the help of diagram and models of sun, moon and earth, (b) Explain how special glasses which will protect us from UV rays, and excess heat can be used to observe eclipses safely.
2. Also inform that eclipses are natural phenomena and can be predicted in advance.

(c) Various eclipses and the conditions during that period.

There are two eclipses:

1. Solar eclipse
2. Lunar eclipse

1. Solar eclipse:

There are two types of Solar eclipse, (a) Total solar eclipse (b) Partial solar eclipse

a. Total solar eclipse: In total solar eclipse, the moon is between the sun and the earth and the moon shadow covers the sun disc fully by perfect alignment. The part of the earth that lies in the umbra of the moon experiences total solar eclipse.

b. Partial solar eclipse: In partial solar eclipse, the moon is between the sun and the earth and the moon shadow does not cover the sun disc fully, because of imperfect alignment. The part of the earth that lies in the penumbra of the moon experiences partial solar eclipse.

c. The solar eclipse occur on a new moon day.

d. Solar eclipse last for few minutes.

2. Lunar eclipse:

There are two types of Lunar eclipse, (a) Total lunar eclipse (b) Partial lunar eclipse

a. Total lunar eclipse: In total lunar eclipse, the earth comes in between the sun and the moon and the earth's shadow covers the moon's surface fully because of perfect alignment.

b. Partial lunar eclipse: In partial lunar eclipse, the earth's shadow does not cover the moon's surface fully because of imperfect alignment. A small part of the moon's surface is covered by umbra part of earth's shadow.

c. The lunar eclipse occurs on a full moon night.

d. Lunar eclipse last for few hours.

8. Explain the difference:

(a) Point sources and Extended sources.

Point sources	Extended sources
1. A source of radiation light that is concentrated at a point and considered as having no spatial extension.	1. An extended source is a source consisting of many point sources separated internally.
2. Shadow we get from a point source is dark called umbra.	2. Shadow we get from extended source of light is faint called penumbra and dark called umbra

(b) Umbra and Penumbra.

Umbra	Penumbra
1. The umbra is a central core of darkness which tapers away.	1. The penumbra is an outer cone of partial shadow which diverges, instead of tapering.
2. Umbra is shadow formed from a point source of light.	2. It is the shadow formed from the extended source of light.
3. It is dark	3. It is faint
4. Umbra is the area of total shadow	4. Penumbra is the area of partial shadow.